

Wahab Sohail

AI Engineer | Conversational AI and Voice Systems

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PROFILE SUMMARY

AI Engineer focused on **conversational AI** and low-resource **Urdu voice systems**. Hands-on experience across **LLM fine-tuning**, **Retrieval-Augmented Generation**, **Coqui TTS**, **Kaldi-based speech recognition**, embeddings, and vector databases. Builds **LangGraph agents** and **FastAPI services**, evaluates grounded responses for factual consistency and task adherence, and deploys AI applications across cloud environments and edge hardware such as **Raspberry Pi 5**.

EDUCATION

National University of Sciences and Technology (NUST)

2022 – 2026

Bachelor of Engineering in Computer Engineering | Islamabad, Pakistan

- Relevant Coursework: Large Language Models, Deep Learning, Machine Learning, Deep Generative Models, Computer Vision, and Digital Signal Processing.

EXPERIENCE

AI Engineer | Blue Group of Companies

- Develop components for an **Urdu conversational voice agent**, spanning local LLM response generation and text-to-speech, with speech-to-text integration in progress for an end-to-end voice pipeline.
- Apply **Coqui TTS** with ElevenLabs audio data to Urdu speech synthesis and investigate natural, practical voice output for domain-specific conversational use cases.
- Fine-tune domain-specific LLMs with **Hugging Face Transformers** and **PEFT/LoRA** on English and Roman Urdu conversations; test generated outputs for factual consistency, directness, and domain adherence.
- Integrate **Ollama-hosted LLMs** and local embedding models through OpenAI-compatible interfaces, resolving backend, agent-configuration, and retrieval integration issues.
- Build and test **RAG pipelines** that retrieve relevant knowledge-base context for grounded Urdu responses.

PROJECTS

SpeakWell – Urdu Phoneme Recognition for Speech Rehabilitation

Final Year Project

Python, C++, Kaldi, HMM Acoustic Models, OpenBLAS, Kivy, Raspberry Pi 5

- Built a **phoneme-level Urdu ASR pipeline** from raw pediatric speech recordings to labelled datasets, addressing the shortage of annotated data for low-resource speech recognition.
- Reimplemented **cross-lingual transfer learning** using English and Persian acoustic models and designed controlled experiments to compare recognition performance across different training-data regimes.
- Deployed the recognizer on **Raspberry Pi 5** using OpenBLAS and hardware-specific build configurations; developed a Kivy interface for live phoneme feedback and rehabilitation progress tracking.

OrgMind – Multi-Agent Research and Question Answering

Python, LangGraph, Groq, Sentence Transformers, Supabase/pgvector, FastAPI, Next.js

- Orchestrated **LangGraph agent workflows** for structured question answering over heterogeneous web data, grounding LLM responses in retrieved source context through a complete **RAG pipeline**.
- Implemented sentence-transformer embeddings, document chunking, and **pgvector retrieval** behind a FastAPI backend; refined retrieval strategies to improve context relevance.
- Delivered the end-to-end application with a Next.js interface and deployed it through Vercel and cloud infrastructure.

Customer Support Ticket Resolution Agent

Python, LangGraph, FAISS, RAG, Docker

- Built a **LangGraph ticket-resolution workflow** combining ticket triage, FAISS knowledge retrieval, and LLM-generated responses grounded in relevant support documentation.
- Containerized the application with **Docker** to make the agent runtime and its dependencies reproducible across development and deployment environments.

SKILLS AND TECHNOLOGIES

LLMs and Agents:	Hugging Face Transformers, PEFT, LoRA, LangGraph, LangChain, Ollama, Groq, Prompt Engineering
Voice AI:	Coqui TTS, Kaldi, HMM Acoustic Models, Urdu ASR, Speech Dataset Preparation, ElevenLabs Audio Data
RAG and Retrieval:	Embeddings, Chunking, Reranking, Sentence Transformers, FAISS, pgvector, ChromaDB
ML and Deep Learning:	PyTorch, TensorFlow, Scikit-learn, Transformers, Model Training, Fine-Tuning, Evaluation
Languages:	Python, C++, SQL, TypeScript, Shell Scripting
Backend and APIs:	FastAPI, REST APIs, OpenAI-Compatible APIs, PostgreSQL, Supabase
Infrastructure and Tools:	Docker, Linux, Git, CUDA, OpenBLAS, Raspberry Pi, Vercel
Research Skills:	Experiment Design, Paper Reimplementation, Cross-Lingual Transfer Learning, Model Evaluation, Technical Documentation

LEADERSHIP

President | COMPPEC, NUST

2025 – 2026

- Led the planning and delivery of a **national-level computing event** with more than 900 participants; coordinated cross-functional teams, secured corporate sponsorships, and managed budgeting and logistics.